

PROJECT REPORT

ServiceNow-Based Employee Laptop Request Application

Platform: ServiceNow (Scoped Application)

Module Type: Service Catalog / Record Producer / Flow Designer

1. Introduction

1.1 Project Overview

The Employee Laptop Request Application is built entirely on the ServiceNow platform as a custom scoped application (scope prefix x_2065290_rr_app_0, internally called the RR application). Rather than following a traditional MERN-stack build, every layer of the solution — frontend, backend, and database — is implemented using native ServiceNow constructs: a Service Catalog / Record Producer for the user interface, Flow Designer and Business Rules for server-side logic, and a custom ServiceNow table for data storage.

This approach allows the application to be delivered, secured, and deployed using the same governance model as the rest of the ServiceNow instance, while still meeting all the functional goals of a standalone laptop-request system.

1.2 Purpose

The purpose of the application is to let employees raise a request for a new laptop directly from a ServiceNow catalog form, instead of relying on an ad-hoc, email-based process. Once submitted, the request is automatically routed to the IT Manager group for approval, and both the requester and the approving manager are notified by email at each stage of the process. The goal is to digitize and standardize laptop provisioning so that every request is tracked, auditable, and resolved consistently.

2. Ideation Phase

2.1 Problem Statement

Employees who need a new laptop currently have no standard channel to raise the request — approvals happen informally over email or chat, with no consistent record of who approved what, when, or why. This makes it hard for IT to track pending requests, prevents managers from seeing an approval history, and leaves no safeguard against an employee submitting multiple requests in a short period.

"How might we give employees a simple, trackable way to request a laptop, while giving IT managers a structured approval workflow with full visibility into request history?"

2.2 Empathy Map Canvas

Looking at the problem from the perspective of the two primary users of the system:

- Employee (Requester) — Says: “I just need a laptop, why is this so hard to ask for?” Thinks: worried the request will get lost in someone’s inbox. Does: sends an email or chat message and waits. Feels: uncertain about status and turnaround time.
- IT Manager (Approver) — Says: “I don’t have a single place to see all pending requests.” Thinks: concerned about approving duplicate or unnecessary requests. Does: manually cross-checks emails before approving. Feels: overloaded by ad-hoc approval messages.

2.3 Brainstorming

Candidate features considered during ideation, several of which were carried into the final solution:

- A catalog-based intake form so employees raise requests through a familiar, standardized channel.
- A dedicated data table to store every request with its current status.
- An automated approval workflow that routes requests to the IT Manager group instead of a named individual.
- Conditional handling based on the type of laptop requested (e.g., Basic vs Advance).
- Automatic email notifications at each stage (submission, approval, rejection).
- A safeguard rule preventing an employee from submitting more than one request per calendar year.

- Role-based access so only authorized employees and managers can create, view, or approve requests.
- A consolidated view showing a request and its full approval history together.

3. Requirement Analysis

3.1 Customer Journey map

The employee's journey through the system follows a simple, linear path:

- 1. Discover — the employee opens the “Employee laptop request” item from the Service Catalog.
- 2. Submit — the employee fills in the Record Producer form (name, email, manager, preferred date, laptop type, reason) and clicks Submit.
- 3. Track — a Laptop Request Form record is created with status “requested”, and the employee can reopen it to see live status.
- 4. Approve/Reject — the request is routed to the IT Manager group; an approver reviews it and approves or rejects it.
- 5. Notify — the Flow updates the record and sends an email to the requester (and manager) reflecting the outcome.
- 6. Review — the requester can see the full approval history on the same form, via the linked Relationship record.

3.2 Solution Requirement

Functional requirements:

- Employees must be able to submit a laptop request with their name, email, manager, preferred date, laptop type, and reason.
- The system must automatically route each request to the IT Manager group for approval.
- The system must send email notifications on approval and on rejection.
- The system must block an employee from submitting a second request in the same calendar year.
- The system must show the approval history alongside the original request record.

Non-functional requirements:

- Security — access to requests must be restricted by role (it employees vs it manager).
- Auditability — every request and its approval trail must be traceable.
- Portability — all customizations must be packaged so the application can be migrated between ServiceNow instances.
- Usability — the intake form should use native ServiceNow UI so it inherits standard responsive layout and accessibility.

3.3 Data Flow Diagram

At a high level, data flows from the employee-facing Record Producer into the Laptop Request Form table, which triggers the Flow. The Flow branches on the requested laptop type, raises an approval for the IT Manager group, and then writes the outcome back to the same record while triggering an email notification.

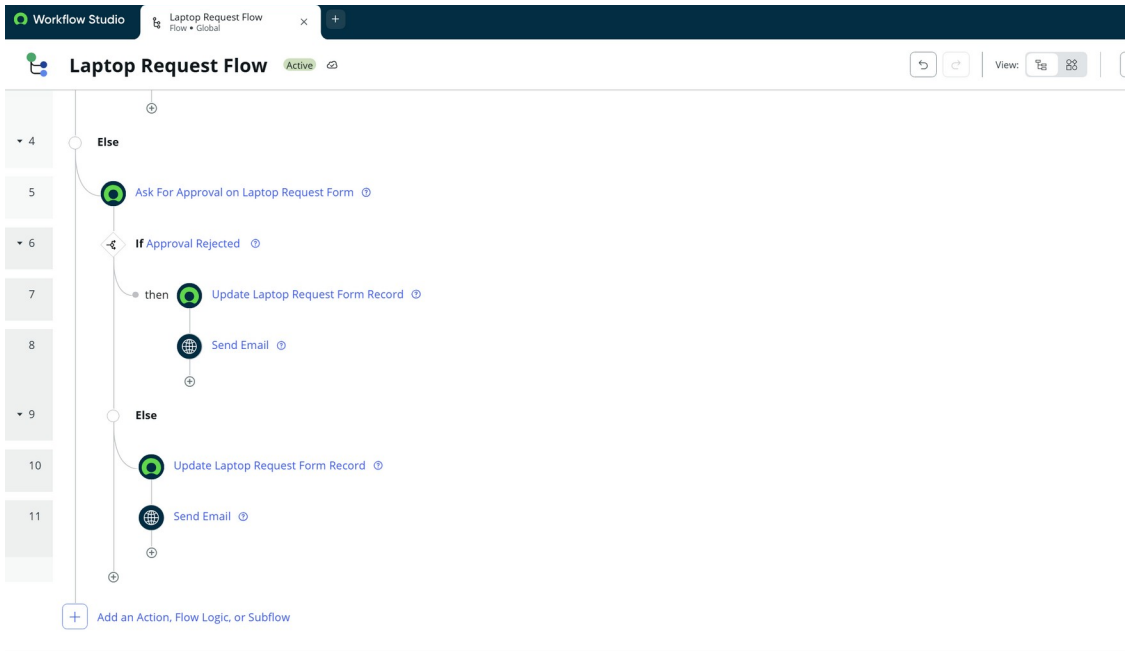


Fig. A — Flow trigger (“Laptop Request Form Created”) and the “Laptop is Basic” condition that checks the Type of Laptop field.

The screenshot displays the 'Laptop Request Form' (LAP0001014) in a web interface. The form contains the following fields: Number (LAP0001014), Requester (Abel Tuter), Employee Email (abel.tuter@example.com), Date (2026-07-20), Short Description, Manager (Sabrina Deppert), Type of Laptop (Advance), Status (requested), and Reason (-- None --). Below the form, there is a table titled 'Approvals' with columns: State, Approver, Comments, Approval for, and Created. The table shows one approval record with State 'Requested', Approver '(empty)', Comments '(empty)', Approval for '(empty)', and Created '2026-07-29 21:41:01'. The table also includes a search bar and a 'Now' button.

Fig. B — Approval branch of the flow: request is routed for approval, then the record is updated and an email is sent depending on approve/reject.

3.4 Technology Stack

Instead of a conventional MERN stack, every layer of the application is a native ServiceNow construct:

Layer	ServiceNow Equivalent
Frontend (React)	Service Catalog Item / Record Producer + native Form/List UI
Backend (Node.js / Express)	Flow Designer flow, Business Rules, Relationship scripts
Database (MongoDB)	Custom table: u_laptop_request_form (and related

Layer	ServiceNow Equivalent
	sys_approval_approver)

4. Project Design

4.1 Problem - Solution Fit

The core problem — an untracked, email-based laptop request process — is solved by moving the entire flow onto a platform employees and managers already use daily. Because ServiceNow natively provides catalog forms, workflow automation, and role-based access control, the same platform that hosts the request also enforces the approval process and security, removing the need for a separate application or manual tracking.

4.2 Proposed Solution

The proposed solution is a single scoped ServiceNow application containing:

- A Record Producer (“Employee laptop request”) presenting a simple catalog form to employees.
- A custom table, Laptop Request Form (u_laptop_request_form), storing every request.
- A Flow Designer workflow that asks the IT Manager group to approve or reject each request, branching on the Type of Laptop.
- Automatic email notifications on approval and rejection.
- A Business Rule that blocks more than one request per employee per calendar year.
- Two custom roles (it employees, it manager) and two groups (it employees, it managers), enforced through Access Control Rules.
- A Relationship record surfacing the approval history directly on the Laptop Request Form.

4.3 Solution Architecture

All components — table, record producer, flow, business rule, roles, groups, and ACLs — live inside a single custom scoped application called RR application (scope prefix x_2065290_rr_app_0), created from ServiceNow Studio. Building in an application scope keeps the tables and logic isolated from the Global scope and makes the whole application easy to move between instances as one Update Set.

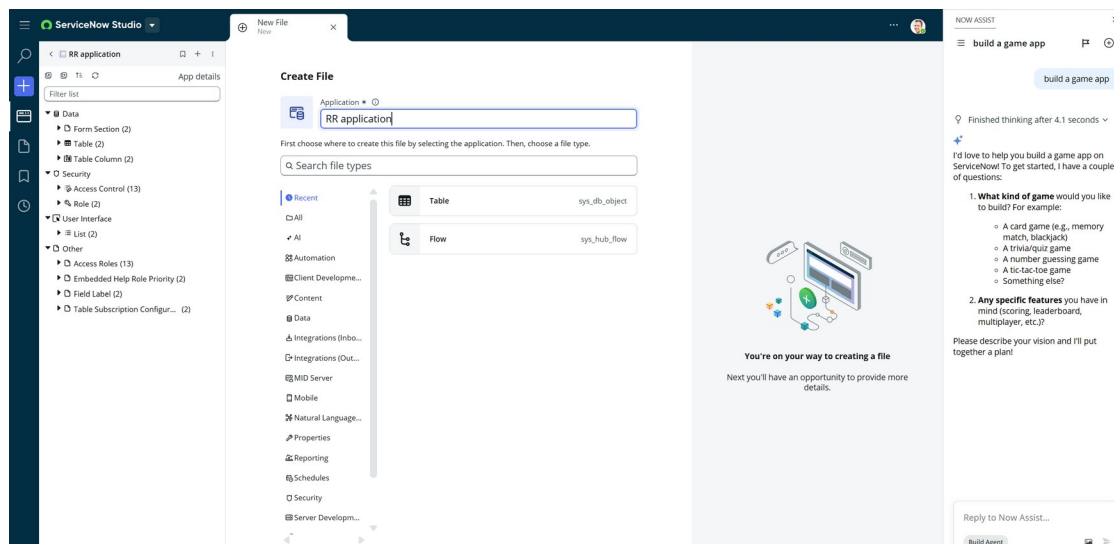


Fig. C — Creating a new file inside the RR application in ServiceNow Studio.

The Relationship record joins the Laptop Request Form table to the Approval (sysapproval_approver) table using a refineQuery script, so every approval raised for a request shows up as a related list directly on the form.

Fig. D — The “Laptop request” Relationship record joining the Approval table to the current Laptop Request Form record.

5. Project Planning & Scheduling

5.1 Project Planning

The build was organized around the main components of the scoped application, created inside ServiceNow Studio and Workflow Studio:

Component	Name / Table	Purpose
Table	Laptop Request Form (u_laptop_request_form)	Stores every laptop request record
Record Producer	Employee laptop request	Catalog form used by employees to raise a request
Flow	Laptop Request Flow	Approval routing and notification logic
Business Rule	Laptop request BR	Blocks duplicate requests in the same year
Relationship	Laptop request	Shows related Approval records on the form
Roles	it employees, it manager	Custom security roles
Groups	it employees, it managers	Assignment/approval groups
Access Controls	4 ACLs on Laptop Request Form	Read / Create / Write security

Before development, the account used for building the application needed the security_admin role elevated, since creating or modifying Access Control Rules requires elevated privilege.

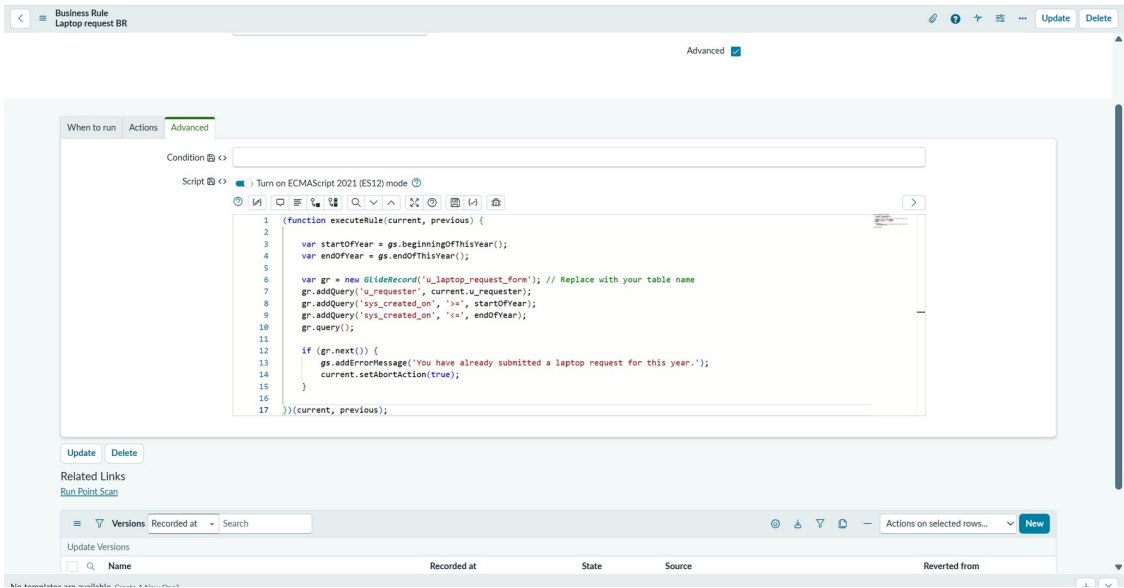


Fig. E — Elevating to the security_admin role from the user profile menu, required before ACLs can be created or edited.

All customizations were captured in a single Update Set (“HR Onboarding Application”), which can be exported to XML and imported into another instance, previewed, and committed — making deployment a repeatable, four-step process.

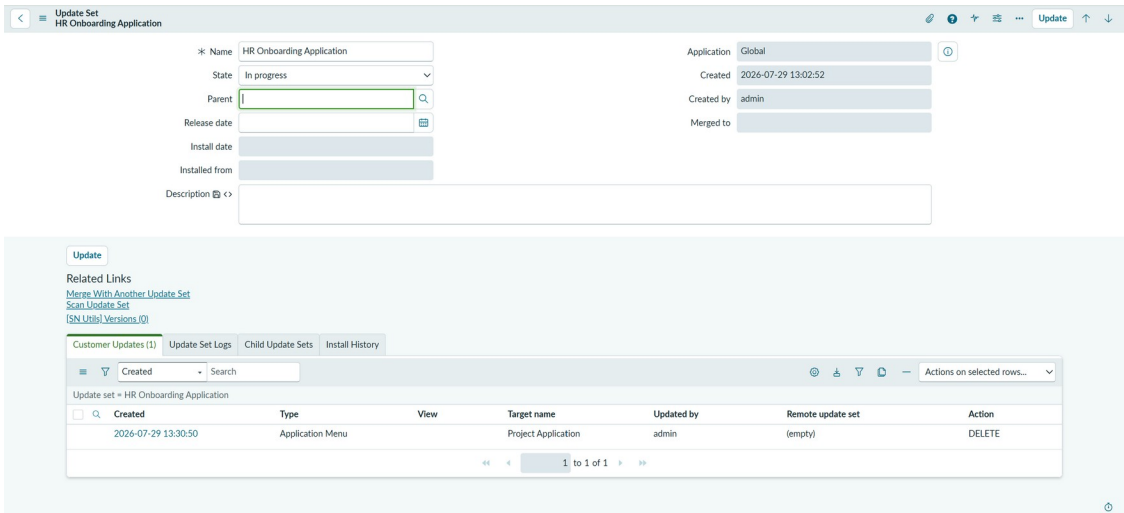


Fig. F — The Update Set in progress, showing the customer updates captured for this project.

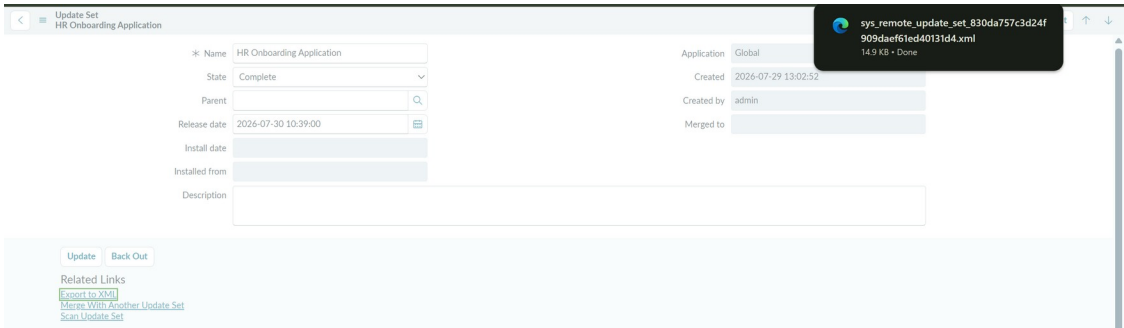


Fig. G — Exporting the completed Update Set to an XML file for migration to another ServiceNow instance.

6. Functional and Performance Testing

6.1 Performance Testing

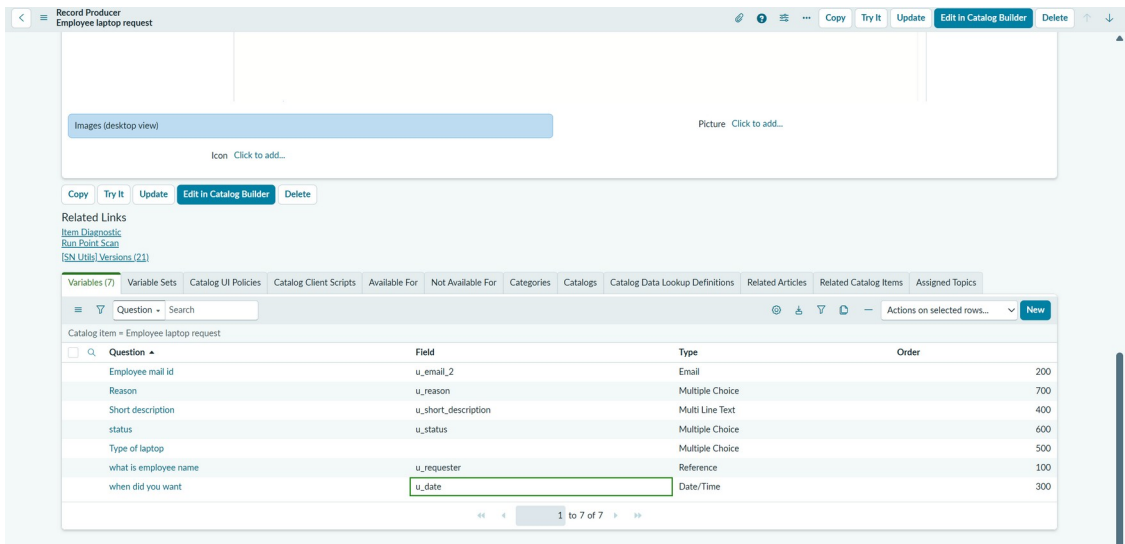
Testing was carried out manually within the ServiceNow instance, covering five areas:

- Form submission testing — submitting the Employee laptop request form with valid and missing field values to confirm mandatory fields are enforced.
- Duplicate-request testing — submitting a second laptop request for the same employee within the same year, confirming the Laptop request BR Business Rule blocks it with the expected error message.
- Flow testing — creating requests with different Type of Laptop values to confirm the “Laptop is Basic” condition branches correctly, and approving/rejecting the resulting approval record to confirm the status updates and the correct email is sent.
- Security testing — logging in as a user with only the “it employees” role to confirm they can create and view their own requests but cannot access manager-only functions, and as a user with the “it manager” role to confirm they can read requests routed to them for approval.
- Update Set validation — exporting the Update Set to XML and re-importing it into a clean instance to confirm every component (table, record producer, flow, business rule, relationship, roles, groups, ACLs) migrates correctly.

7. Results

7.1 Output Screenshots

The following screenshots capture the working application end to end — from the Record Producer fields through to the security model enforcing access.



The 7 variables/fields defined on the Employee laptop request Record Producer, mapped to columns of the u_laptop_request_form table.

Relationship
Laptop request

Back

Name

Laptop request

Advanced

Application

Global

Applies to table

Laptop Request Form [x_2065290_rt_app...

Queries from table

Approval [sysapproval_approver]

Run Query Diagnostics

Update

Delete

This script refines the query in current that will populate the related list. For more information about it, its parameters and control variables, see [the documentation](#). See also the article about the [recommended form of the script](#).

Query with

Turn on ECMAScript 2021 (ES12) mode

```
1 (function refineQuery(current, parent) {
2
3   // Add your code here, such as current.addQuery(field, value)
4   current.addQuery('source_table', parent.getTableName());
5   current.addQuery('document_id', parent.sys_id);
6
7 })(current, parent);
```

Run Query Diagnostics

Update

Delete

Related Links

[Run Point Scan](#)

[\[SN Utils\] Versions \(1\)](#)

Advanced script of the “Laptop request BR” Business Rule, using `gs.beginningOfThisYear() / gs.endOfThisYear()` and `current.setAbortAction(true)` to block a duplicate submission.

Group
it employees

Update

Delete

Name

it employees

Manager

Group email

Parent

Description

Employee members in it companies

Update

Delete

Roles (1)

Group Members (10)

Groups

Group = it employees

User

Search

Abel Tuter

Benchmark Scheduler

Allyson Gillisple

Chase Furler

Callie Leboeuf

Berta Karczewski

Celia Slavin

Cherie Fuhri

Alleem Mottern

Antony Alidis

1 to 10 of 10

The blank Laptop Request Form as presented to the employee before submission.

Role
it manager

Name

it manager

Application

Global

Description

manager of it company

Elevated privilege

☐

Update

Delete

Contains Roles

Applications with Role

Modules with Role

Custom Tables

for text

Search

Role = it manager

Contains

No records to display

A submitted Laptop Request Form record showing the requester, manager, laptop type and the related Approval record in “Requested” state.

Group
it managers

Job to add or remove role(s) from user(s) of group has been queued

Name

it managers

Group email

Manager

Parent

Description

who approves the request

Update

Delete

Roles (1)

Group Members (8)

Groups

User

Search

Group = it managers

User

Byron Fortuna

Ashley Leonesio

Bryan Rovell

Cary Mcamey

Bow Ruggeri

Bette Barcelona

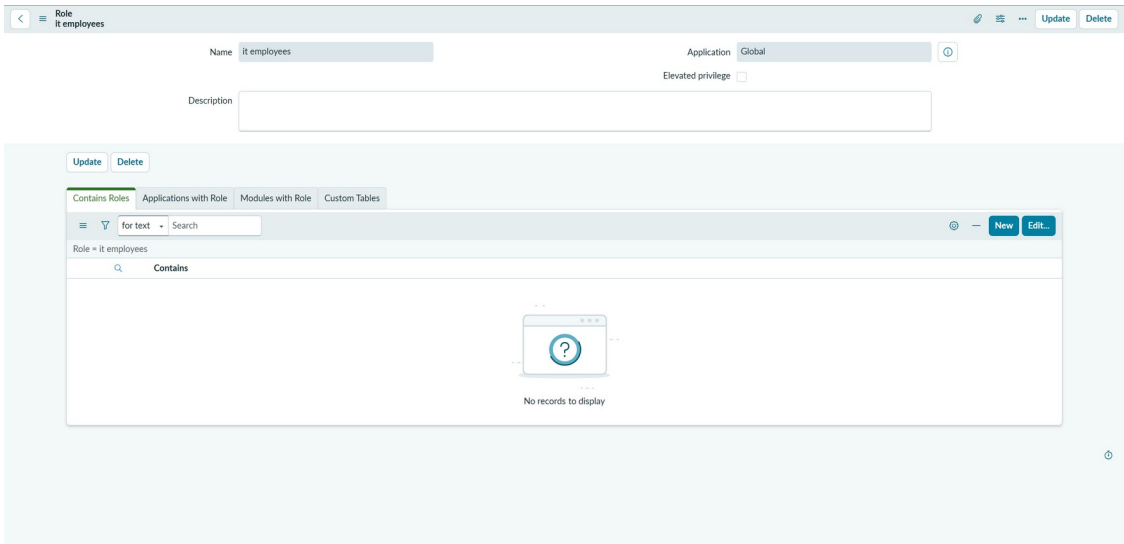
Benchmark Scheduler

Candice Bruckman

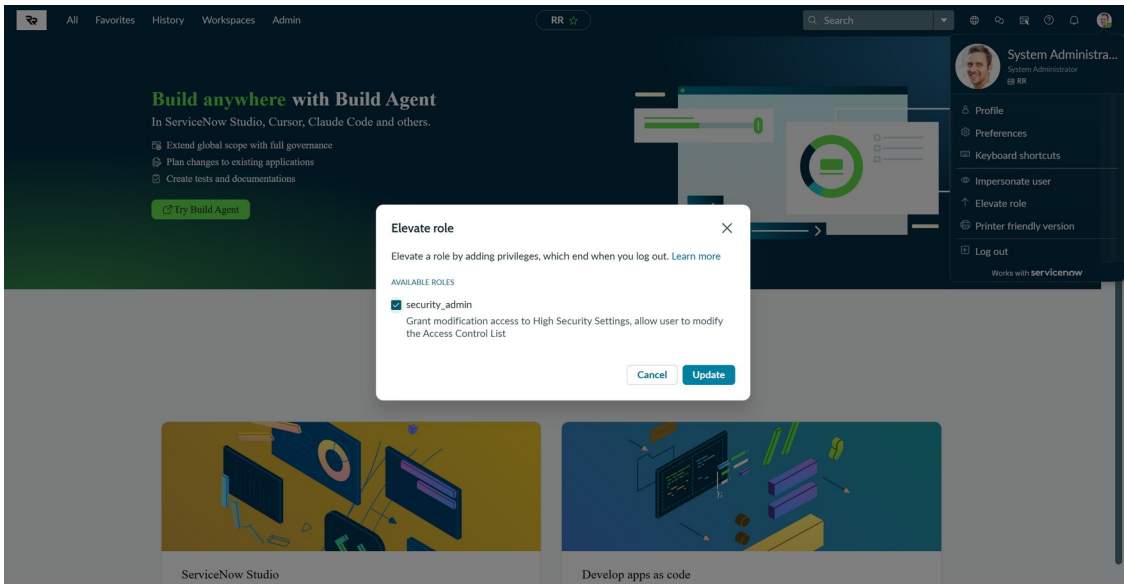
1 to 8 of 8

The custom “it employees” role, granted to staff who are allowed to submit laptop requests.

9



The custom “it manager” role, granted to IT managers who approve or reject laptop requests.



The “it managers” group — the approval group used by the Laptop Request Flow.

Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

* Type: record
 * Operation: read
 Admin overrides: ☒
 Decision Type: Allow If
 Protection policy: -- None --
 * Name: Laptop Request Form [x_2065290_rr_app_0_laptop_request_form]
 Description:
 Application: Global
 Active: ☒
 Advanced: ☐

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.
 1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
 2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.
[More Info](#)

Requires role
Role
✗ it manager
+ Insert a new row...

The “it employees” group, representing staff entitled to submit requests.

* Type: record
 * Operation: create
 Admin overrides: ☒
 Decision Type: Allow If
 Protection policy: -- None --
 * Name: Laptop Request Form [x_2065290_rr_app_0_laptop_request_form]
 Description: Allow create for records in x_2065290_rr_app_0_laptop_request_form, for users with role it employees.
 Application: Global
 Active: ☒
 Advanced: ☐

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.
 1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
 2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.
[More Info](#)

Requires role
Role
✗ it employees
+ Insert a new row...

Read Access Control on Laptop Request Form, restricted to the “it manager” role.

Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

* Type: record
 * Operation: read
 Admin overrides: ☒
 Decision Type: Allow If
 Protection policy: -- None --
 * Name: Laptop Request Form [x_2065290_rr_app_0_laptop_request_form]
 Description:
 Application: Global
 Active: ☒
 Advanced: ☐

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.
 1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
 2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.
[More Info](#)

Requires role
Role
✗ it employees
+ Insert a new row...

Create Access Control on Laptop Request Form, restricted to the “it employees” role.

The screenshot shows the 'Access Control' configuration page for the 'Laptop Request Form' (x_2065290_rr_app_0_laptop_request_form). The configuration includes:

- Type:** record
- Operation:** write
- Admin overrides:** checked
- Decision Type:** Allow if
- Protection policy:** None
- Name:** Laptop Request Form [x_2065290_rr_app_0_laptop_request_form]
- Description:** Allow write for records in x_2065290_rr_app_0_laptop_request_form, for users with role it employees.
- Application:** Global
- Active:** checked
- Advanced:** unchecked

Below the configuration, the 'Conditions' section is expanded, showing a message about Access Control Rules and a table for 'Requires role'.

Role
it employees
Insert a new row...

Read Access Control on Laptop Request Form, restricted to the “it employees” role, so requesters can view their own submitted records.

The screenshot shows the 'Workflow Studio' configuration page for the 'Laptop Request Flow' (Flow • Global). The configuration includes:

- TRIGGER:** Laptop Request Form Created
- ACTIONS:** Select multiple
- Condition Label:** Laptop is Basic
- Condition 1:** Trigger - R... → Type of La... is basic
- Logic:** or
- Buttons:** Add another condition set(OR)

Write Access Control on Laptop Request Form, restricted to the “it employees” role.

Summary of Access Control Rules:

Operation	Table	Required Role
create	Laptop Request Form	it employees
read	Laptop Request Form	it employees
write	Laptop Request Form	it employees
read	Laptop Request Form	it manager

8. Advantages & Disadvantages

Advantages

- Runs entirely on the existing ServiceNow instance — no separate frontend/backend server to build, host, or maintain.
- Inherits ServiceNow's native security model (roles, groups, ACLs) instead of custom authentication code.

- Every table automatically exposes a REST Table API, so integrations are possible without hand-written endpoints.
- All customizations are packaged in a single Update Set, making the application easy to migrate between instances.
- The approval history is visible directly on the request record via the Relationship record, giving full auditability.

Disadvantages

- The Type of Laptop variable on the Record Producer has no backing field mapped on the table, so its value is not currently queryable in reports.
- The duplicate-request check only looks at the calendar year — a rejected employee cannot raise a new request until the following year.
- Email notifications use a generic “Send Email” flow action; sender address and templates are not yet finalized or branded.
- The Manager field is a free-text reference lookup and is not auto-populated from the requester's manager record.

9. Conclusion

The ServiceNow-Based Employee Laptop Request Application replaces an ad-hoc, email-driven process with a standardized catalog form, an automated approval flow, and role-based security — all built natively on the ServiceNow platform. Because the frontend, backend, and database layers all live inside a single scoped application, the solution requires no external infrastructure, inherits the platform's built-in security and REST API, and can be packaged into one Update Set for straightforward deployment to other instances. Manual testing across form submission, duplicate-request handling, flow branching, and role-based security confirmed the application behaves as intended, while a small set of known issues (unmapped laptop-type field, year-locked duplicate check, generic email templates, and a non-auto-populated manager field) have been identified for future refinement.

10. Future Scope

- Auto-populate the Manager field from the requester's sys_user.manager record.
- Add a Service Portal widget so employees can track the live status of their request without opening the platform UI.
- Extend the Flow to integrate with Asset Management so an approved request automatically creates a hardware asset assignment task.
- Add a Performance Analytics / reporting dashboard summarizing requests by laptop type, status, and turnaround time.
- Allow a rejected request to be resubmitted with manager comments instead of blocking further requests until the next calendar year.
- Add mobile-friendly UI Actions so managers can approve or reject requests directly from the ServiceNow mobile app.

11. Appendix

Source Code (if any)

As a ServiceNow scoped application, there is no separate source-code repository in the traditional sense — the table, Record Producer, Flow, Business Rule, Relationship, roles, groups, and ACLs are all captured together in the “HR

Onboarding Application” Update Set (exportable to XML) described in Section 5.1. The Business Rule's server-side script is shown in Fig. under Section 7.1.

Dataset Link

Not applicable — the application uses a live ServiceNow table (u_laptop_request_form) populated by user submissions rather than a static dataset.

GitHub & Project Demo Link

[Add GitHub repository link here, if the Update Set XML is version-controlled]

[Add project demo / walkthrough video link here]